

component under test, which then can be compared with the defined specifications.

Most crucial part of this process is generating an accurate circuit model of the PDN network. Any tool is as good as its input. In this case, SPICE simulator output results are as accurate as the circuit model used to generate the same.

Circuit model generation of the PDN under test is a two-way process. At first, frequency domain or s-parameter model of the PDN network is generated using 3D Finite Element Method based solvers. Once the s-parameter model is generated and verified for passivity and casualty, a circuit model can be generated from s-parameter model through macro-modelling based software tools.

Fig. 7 displays an example of time domain voltage and current profiles for any PDN under analysis. A piece-wise linear current source connected at load terminal is used as a stimulus to excite the PDN. Here, the stimuli current rises from 17A to 37A in 1 μ s. In real work application, this type of load demand can be attributed to CPU operating in turbo mode of operation, that is, increasing its operating frequency to execute the workload.

With rise in current demand of the load, voltage at its input drops to a minimum level, governed by PDN impedance Z, and rises back to the steady-state level governed by its DC resistance. At about 10 μ s, the current demand falls suddenly, which could be attributed to end of workload execution and CPU reducing its frequency of operation. Due to sudden decrease in the load demand, voltage at the devices overshoots and hits a maximum value before settling back to the steady state.

The main aim of the entire exercise of power integrity analysis is to ensure minimum and maximum voltage levels at device input

measured in simulation are within device operating range for reliable performance, and the simulation setup and results match closely with actual testing conditions and results.

Fig. 8 shows system power integrity analysis workflow. It starts with sanity check of system receivables followed by DC performance check. With DC criterion met, analysis flow moves to frequency domain analysis. In frequency domain analysis, impedance profile Z of the system is compared against Z_{Target} . If $Z < Z_{\text{Target}}$ for system range of frequency operation, the analysis moves to the final stage of transient check or time domain analysis.

For time domain analysis, PDN circuit model is generated first followed by SPICE deck building. Once the computed voltage droop and overshoot meets the design specifications, the power integrity analysis is concluded with high confidence design readiness.

Conclusion

With modern-day complex multi-chipset systems, power delivery designs are becoming challenging and more critical than ever. A lot of root-cause failures in faulty systems can be somehow related to the power delivery. With power integrity analysis workflow, this process can be simplified. If it sounds too good to be true, let's clarify that it does amount to considerable work and investment during the design phase of the product. However, it ensures smooth design validation, reliable product performance, and better customer experience, which at the end translates to better sales and profits, which are the ultimate goals for any organisation. **EFY**

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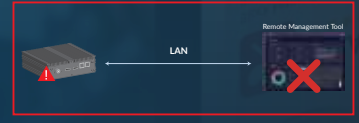
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